

Name: .....

Maths Class: .....

**SYDNEY TECHNICAL HIGH SCHOOL**  
(Est. 1911)



**Year 11**

**Mathematics Extension 1**  
**May 2013**

*Time allowed: 70 min*

***Instructions:***

- Write your name and class at the top of this page.
- These questions must be handed in on the *top* of your answers
- Attempt all questions.
- Begin each question on a new page.

*Use only blue or black pen for your answers*

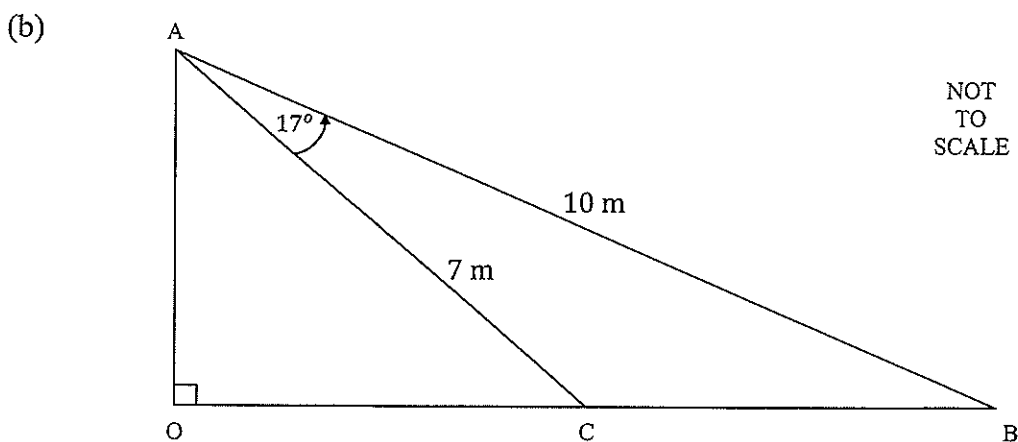
**Total Marks – 60**

**Question 1** (10 marks) Use a SEPARATE page

- (a) Simplify the following expression  $\frac{x^3 - y^3}{x^2 - y^2}$ . 2
- (b) (i) By factorising, simplify  $2^{n+1} + 2^n$ . 1
- (ii) Hence, or otherwise, write  $\frac{2^{1001} + 2^{1000}}{3}$  as a power of 2. 2
- (c) Simplify:  $\frac{10^x + 15^x}{2^5 \times 3^x + 2^{x+5}}$ . 3
- (d) Find the exact value of  $\sin 120^\circ - \tan 210^\circ$ . Express your answer with a rational denominator. 2

**Question 2** (10 marks) Use a SEPARATE page.

- (a) The sum of the interior angles of a regular polygon is  $3960^\circ$ .
- (i) How many sides does the polygon have? 1
- (ii) Find the size of each interior angle. 1
- (iii) Hence or otherwise find the size of the exterior angle. 1



- (i) Find the area of  $\triangle ABC$  to 2 significant figures. 2
- (ii) Find the length of BC to 2 significant figures. 2
- (iii) Find the length of BO to 2 significant figures. 3

**Question 3** (10 marks) Use a SEPARATE page.

(a) Find the *exact* solutions of  $x + 8 = \frac{6}{x}$  3

(b) (i) Draw the graph of  $y = |x - 1|$  and  $y = x + 3$  on the same axes. 2

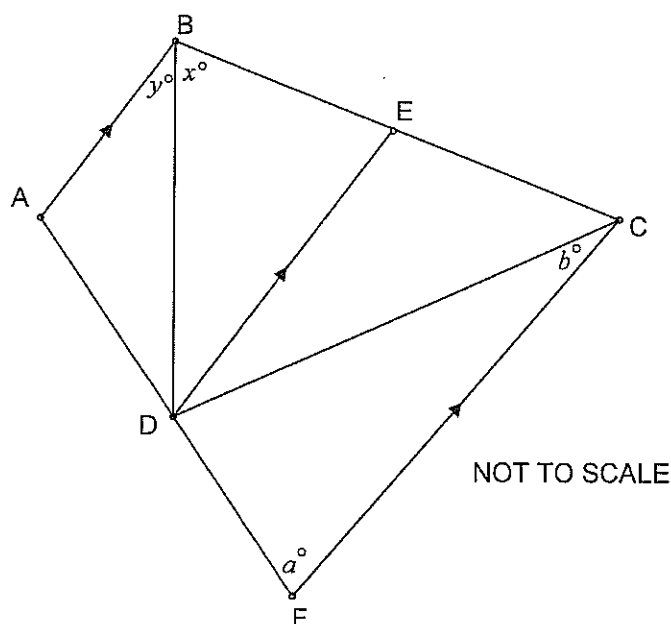
(ii) Hence or otherwise solve  $|x - 1| > x + 3$ . 2

(c) Solve for  $x$ : 3

$$\frac{4x - 1}{x + 2} \geq 3$$

**Question 4** (10 marks) Use a SEPARATE page.

(a) In quadrilateral ABCD,  $AB = AD$ ,  $CB = CD$  and FC is parallel to AB and DE.



(i) Show that  $a = 2y$ , giving reason(s). 3

(ii) Show that  $b = x - y$ , giving reason(s). 2

(b) (i) Use the method of grouping in pairs to factorise fully 2

$$3x^3 + 3x^2 - x - 1.$$

(ii) Hence or otherwise solve 3

$$3\tan^3\theta + 3\tan^2\theta - \tan\theta - 1 = 0 \text{ for } 0 \leq \theta \leq 180^\circ.$$

**Question 5** (10 marks) Use a SEPARATE page.

- (a) For the function  $f(x) = \frac{9}{9-x^2}$
- (i) Giving reasons, is the function odd, even or neither? 1
  - (ii) Find the equation(s) of the asymptotes. 2
  - (iii) Using a ruler, sketch the graph of  $y = f(x)$ , showing all key features. 3
  - (iv) Hence, or otherwise, state the domain and range of the function. 1
- (b) If  $3 \cos \theta + 2 = 0$  and  $\tan \theta > 0$ , what is the exact value of  $\sin \theta$ ? 3

**Question 6** (10 marks) Use a SEPARATE page.

- (a) Jade is on a ship and observes two lighthouses on the shore. The lighthouse at Addison Head has a bearing of  $224^\circ$  from the ship. The lighthouse at Blake Beach has a bearing of  $195^\circ$  from the ship and  $165^\circ$  from Addison Head. The lighthouses are 3.4 km apart.
- (i) Draw a diagram showing all necessary information. 2
  - (ii) What is the distance of Jade's ship from the Addison Head lighthouse (1 decimal place)? 2
- (b) Prove 3
- $$\frac{(1 + \tan^2 \theta) \cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$$
- (c) If  $2^a + 3^b = 17$  and  $2^{a+2} - 3^{b+1} = 5$ , find the values of  $a$  and  $b$ . 3

**End of test**

Q1

$$\begin{aligned}
 (a) & \frac{(x-y)(x^2+xy+y^2)}{(x-y)(x+y)} \\
 &= \frac{x^2+xy+y^2}{(x+y)}
 \end{aligned}$$

$$\begin{aligned}
 (b)(i) & 2 \times 2^n + 2^n \\
 &= 2^n(2+1) \\
 &= \underline{\underline{3 \times 2^n}}
 \end{aligned}$$

$$\begin{aligned}
 (ii) & \frac{2 \times 2^{1000} + 2^{1000}}{3} \\
 &= \frac{3 \times 2^{1000}}{3} = \underline{\underline{2^{1000}}}
 \end{aligned}$$

$$\begin{aligned}
 (c) & \frac{2^x 5^x + 3^x 5^x}{2^5 \times 3^x + 2^5 \times 2^x} \\
 &= \frac{5^x(2^x + 3^x)}{2^5(3^x + 2^x)} \\
 &= \underline{\underline{\frac{5^x}{2^5}}}
 \end{aligned}$$

$$\begin{aligned}
 (d) & \sin(180-60) - \tan(180+30) \\
 &= \sin 60 - \tan 30 \\
 &= \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{3} \\
 &= \frac{3\sqrt{3} - 2\sqrt{3}}{6} = \underline{\underline{\frac{\sqrt{3}}{6}}}
 \end{aligned}$$

Q2

$$\begin{aligned}
 (a) (i) & (3960 \div 180) + 2 = \underline{\underline{24}} \\
 (ii) & 3960 \div 24 = \underline{\underline{165}} \\
 (iii) & 180 - 165 = \underline{\underline{15}}
 \end{aligned}$$

$$\begin{aligned}
 (b)(i) & A = \frac{1}{2} \times 10 \times 7 \times \sin 17 \\
 &= \frac{1}{2} \times 10 \times 7 \times 0.2924 \\
 &= \underline{\underline{10 \text{ m}^2 \text{ (2 s.f.)}}}
 \end{aligned}$$

$$\begin{aligned}
 (ii) & BC^2 = 10^2 + 7^2 - 2 \times 10 \times 7 \times \cos 17 \\
 &= 15.117 \dots \\
 \therefore BC &= \underline{\underline{3.9 \text{ m}}}
 \end{aligned}$$

$$(iii) \frac{\sin B}{7} = \frac{\sin 17}{3.9}$$

$$\begin{aligned}
 \sin B &= 0.5247692 \dots \\
 B &= \underline{\underline{31.6527^\circ}}
 \end{aligned}$$

$$\therefore \frac{OB}{10} = \cos 31.6527$$

$$\begin{aligned}
 OB &= 10 \cos 31.6527 \\
 &= \underline{\underline{8.5 \text{ m}}}
 \end{aligned}$$

Q3

$$(a) \quad x^2 + 8x = \frac{6 \times x}{x}$$

$$x^2 + 8x = 6$$

$$x^2 + 8x - 6 = 0$$

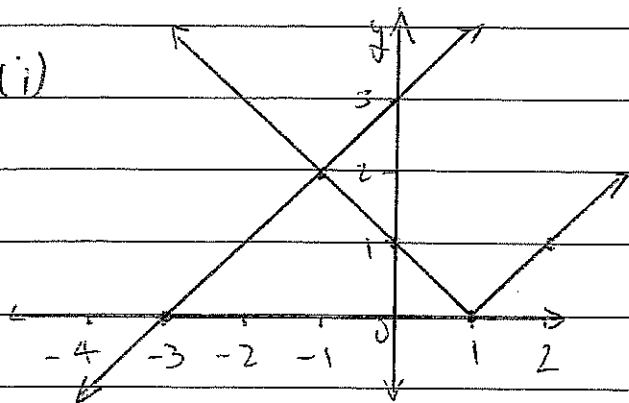
$$x = \frac{-8 \pm \sqrt{8^2 - 4 \times 1 \times (-6)}}{2 \times 1}$$

$$= \frac{-8 \pm \sqrt{88}}{2}$$

$$= \frac{-8 \pm 2\sqrt{22}}{2}$$

$$x = \underline{\underline{-4 \pm \sqrt{22}}}$$

b)(i)

(ii) when  $x < -1$ 

$$c) \quad \frac{4x-1}{x+2} \times (x+2)^2 \geq 3(x+2)^2$$

$$(4x-1)(x+2) \geq 3(x+2)^2$$

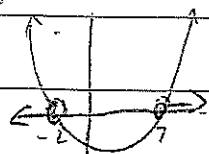
3c) continued.

$$(4x-1)(x+2) - 3(x+2)^2 \geq 0$$

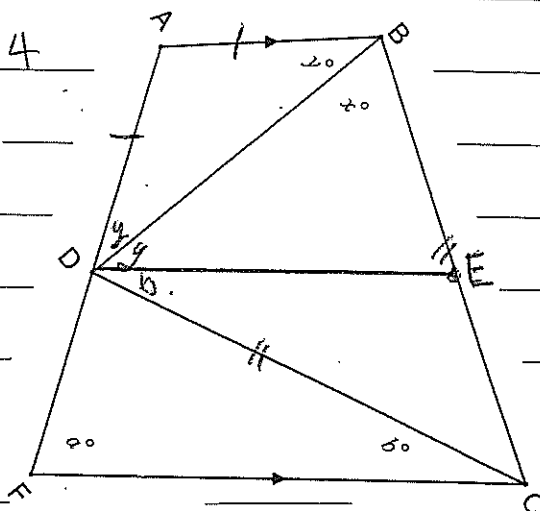
$$(x+2)[4x-1-3(x+2)] \geq 0$$

$$(x+2)(x-7) \geq 0$$

$$\therefore \underline{\underline{x \leq -2 \text{ \& } x \geq 7}}$$



Q4

(i)  $AD = AB$  (given)
 $\angle ABD = \angle ADB$  (equal angles of an isosceles  $\triangle ABD$ )

 $\angle BDE = y$  (alternate angles,  $AB \parallel DE$ )  
 $\therefore x = 2y$  (corresponding angles,  $DE \parallel FC$ )
(ii)  $CB = CD$  (given)
 $\angle CDE = b$  (alternate angles,  $DE \parallel FC$ )  
 $\angle DBC = \angle DCB$  (equal angles of an isosceles  $\triangle CBD$ )

$$b + y = x$$

$$\therefore b = x - y$$

Q4 continued...

$$(b)(i) \quad 3x^2(x+1) - (x+1)$$

$$(x+1)(3x^2-1)$$

$$(2c+1)(\sqrt{3}2c-1)(\sqrt{3}2c+1)$$

ii)  $(\tan \theta + 1)(\sqrt{3} \tan \theta - 1)(\sqrt{3} \tan \theta + 1) = 0 \therefore \sin \theta = -\sqrt{5}$

$$\therefore \tan \theta = -1 \quad \tan \theta = \frac{1}{\sqrt{3}} \quad \tan \theta = -\frac{1}{\sqrt{3}}$$

$$\theta = 135^\circ$$

$$\theta = 30^\circ$$

$$\theta = 150^\circ$$

05

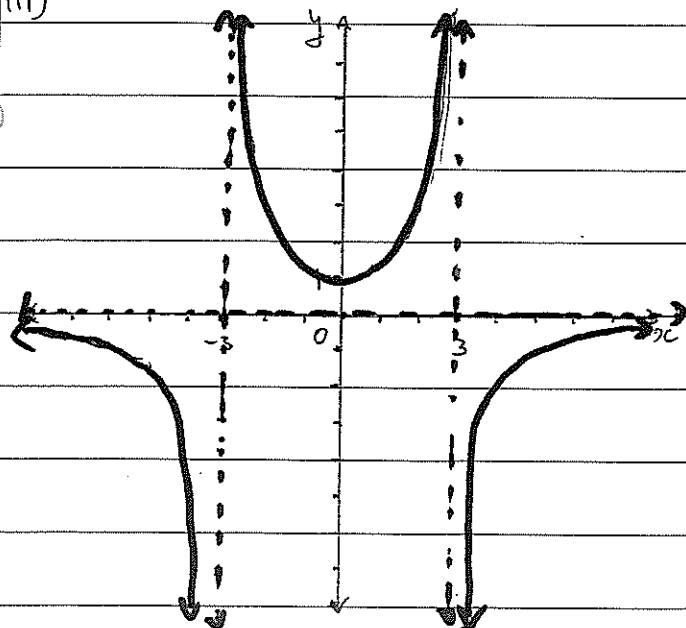
$$(ii) f(-x) = \frac{6}{9 - (-x)^2} = \frac{6}{9 - x^2} = f(x) \quad \therefore \text{even.}$$

(ii)  $9 - x^2 \neq 0$

$$x \neq \pm 3$$

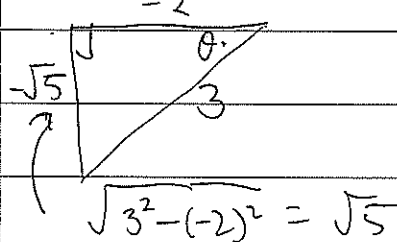
$y \neq 0$

(ii)

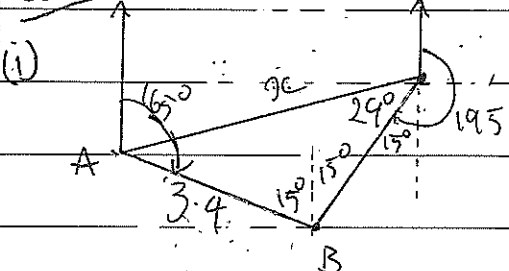


(b)  $\cos \theta = -\frac{2}{3}$   $\cos \theta < 0$   
 $\tan \theta > 0$

$\tan \theta > 0$   
 $\therefore$  3rd quadrant



Q6. N



$$\text{ii) } \frac{x}{\sin 30} = \frac{3.4}{\sin 29}$$

∴  $x = 3.5 \text{ km}$

$$\begin{aligned} \text{(b) } \sec^2 \theta \times \cos \theta &= \frac{1}{\sin^2 \theta} \times \cos \theta = \frac{\cos \theta}{\sin^2 \theta} \\ \text{LHS} &= \frac{1}{\cos^2 \theta} \times \frac{\cos \theta}{\sin \theta} = \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta = \text{RHS} \end{aligned}$$

(c) Let  $x = 2^a$  &  $y = 3^b$

$$x + y = 17 \quad (1)$$

$$1^{a+2} - 3^{b+1} = 5 \Rightarrow 2^2 \times 2^a - 3 \times 3^b = 5$$

$$\therefore 4x - 3y = 5 \text{ (2)}$$

$$3 \times (i) \quad 3x + 3y = 1 \quad (ii)$$

$$\textcircled{3} + \textcircled{2} \quad 70^\circ = 56 \therefore 74 = 8$$

$$8 + 9 = 17$$

$$2^a = 8$$

$$y = 9$$

$$\therefore a = 3$$

$$\therefore 3^b = 9$$

$$\therefore \underline{b=3}$$

iv)  $\mathbb{R}: y \geq 1$  and  $y < 0$   $\mathbb{D}$ : all  $x$ , except

$$00 \neq \pm 3$$